

Load Interconnection Request System Impact Study Report

For Queue Project Q22-2146

Yellow Creek 138 kV
August 21, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2146. The Interconnection Customer proposes a 182.7 MW large load (data center) facility interconnecting at the Yellow Creek 138 kV substation in the Virginia area. The request was received on July 29, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 447 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$723,000.

3. Network Impacts

3.1 Load Deliverability Analysis

For the loss of GENEVA BORDER CITY - DEWALT 138 KV Line, the ELY - NOTTOWAY DIESEL GENERATING FACILITY 138 KV loads to 101.1% of its 222 MVA rating. Project DFAX: 6.6%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
ELY - NOTTOWAY DIESEL GENERATING FACILITY 138 KV	Reconductor line	\$6,392,000	11% (\$723,000)
	Total allocated to project		\$723,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.